

Overview of Study Design and Validation Strategy

1. Cohorts and Reference Data

Discovery cohort: GSE14520 (n=221; OS)
External cohorts: TCGA-LIHC, ICGC-LIRI-JP,
GEO (GSE76427/GSE10143/GSE27150)
Ferroptosis reference: FerrDb V2

2. Candidate Gene Derivation

Differential expression in GSE14520
(limma Bioconductor package)
FerrDb genes filtered by DEG significance
Final modeling candidate set: 125 genes

3. Prognostic Signature Training

Univariate Cox and LASSO-Cox (10-fold CV)
Nine-gene fixed-coefficient risk score
Internal evaluation: KM survival,
time-dependent ROC (1/3/5 years), bootstrap C-index

4. External Validation

Testing in TCGA-LIHC, ICGC-LIRI-JP,
and GEO cohorts
Metrics: C-index, time-dependent AUC,
3-year calibration, IPCW Brier
Robustness: PH checks, RMST,
random-signature benchmark

5. Prediction and Biological Context

Prediction-model context: multivariable Cox,
nomogram, decision-curve analysis
Biological context: immune signatures,
checkpoints, TCGA mutation/CNV/subtypes

6. Interpretation and Next Steps

Findings support HCC risk stratification
and hypothesis generation.
Prospective validation with predefined
thresholds is required.